

FSD2 LAB • EXPERIMENT 04

ExpressJS Database with Mongoose, CRUD, RESTful API

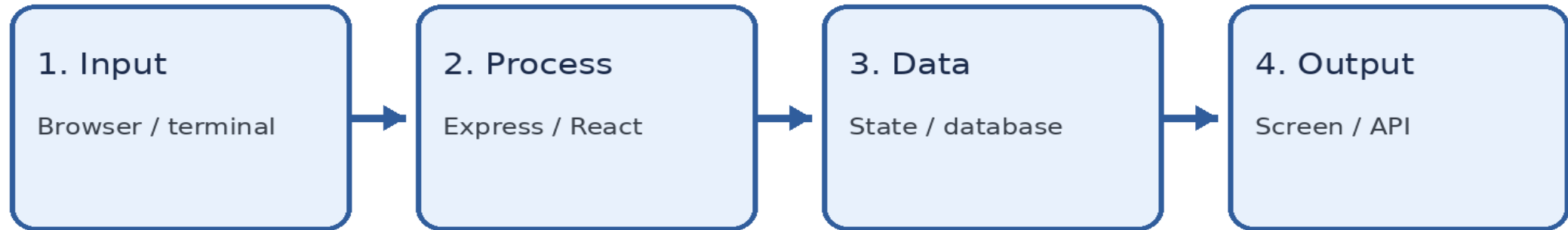
GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Connect MongoDB using Mongoose and perform CRUD operations.
- Develop a single page application using RESTful APIs.

2. Workflow

ExpressJS Database with Mongoose, CRUD, RESTful API



Workflow: input → processing → data/state → output

3.1. Code pathway — server.js

```
1 const express = require('express');
2 const mongoose = require('mongoose');
3 const app = express();
4 const PORT = 3004;
5
6 app.use(express.json());
7
8 const productSchema = new mongoose.Schema({
9   name: String,
10  price: Number
```

3.2. Code pathway — server.js

```
11 });  
12 const Product = mongoose.model('Product', productSchema);  
13  
14 // REST: GET all products.  
15 app.get('/api/products', async (req, res) => {  
16   res.json(await Product.find());  
17 });  
18  
19 // REST: POST a product.  
20 app.post('/api/products', async (req, res) => {
```

4. Runtime walkthrough

Runtime walkthrough — ExpressJS Database with Mongoose, CRUD, RESTful



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?